

E-COMMERCE SALES, CUSTOMER & DELIVERY PERFORMANCE ANALYSIS

1. Project Overview and Objective

This project focuses on analysing e-commerce order data to understand sales performance, customer contribution, product-category performance, purchasing channels, and delivery operations. The workflow starts with raw-data inspection and Python-based data cleaning, followed by feature engineering, exploratory data analysis (EDA), business interpretation, and an interactive Power BI reporting layer.

The main objective is to transform raw e-commerce data into meaningful business insights and present the findings through Python visualizations and an interactive Power BI dashboard.

- Analyse overall revenue and order performance.
- Identify high-performing product categories.
- Evaluate customer-segment contribution to revenue and orders.
- Understand monthly revenue and order trends.
- Analyse payment methods, sales channels, and device usage.
- Evaluate delivery status and shipping performance.
- Develop key insights and business recommendations.
- Create an interactive Power BI dashboard for decision-making.

2. Data Sources

Source Description: <https://huggingface.co/datasets>.

Domain: E-Commerce / Sales Analytics

Dataset Size: 10,000 records and 15 raw columns.

Data Period: 20 April 2024 to 19 April 2025.

The raw dataset contains order, customer, product, pricing, shipping, payment, device, sales-channel and customer-segment information. The data contains missing values and inconsistent categorical entries that were handled during Python preprocessing.

3. Problem Statement

- Analyse overall e-commerce revenue and order performance.
- Identify the highest-revenue product categories.
- Analyse customer-segment contribution to revenue and orders.
- Study monthly revenue and order trends.
- Compare payment methods, sales channels and device types.
- Evaluate delivery status and average shipping duration.
- Identify data-quality issues before analysis.
- Provide recommendations based on the analytical findings.

4. Attribute (Column / Features) Details

Attribute Name	Data Type	Description
order_id	String	Unique order identifier
customer_id	ID	Customer identifier
product_id	ID	Product identifier
category	Categorical	Product category
price	Decimal	Product/order price
quantity	Numeric	Quantity ordered
order_date	Date/Time	Order date
shipping_date	Date/Time	Shipping date
delivery_status	Categorical	Delivery status
payment_method	Categorical	Payment method
device_type	Categorical	Device used for order
channel	Categorical	Sales channel
customer_segment	Categorical	Customer segment

Two additional analytical features were created in Python: $\text{revenue} = \text{price} \times \text{quantity}$ and $\text{shipping_days} = \text{shipping_date} - \text{order_date}$.

5. Tools & Technologies

Tool	Use in Project
Python	Data loading, cleaning, preprocessing and EDA
Pandas	DataFrame operations, missing values, grouping and pivot analysis
NumPy	Numerical operations
Matplotlib	Python analytical visualizations
Google Colab	Python notebook execution and documentation
Power BI	Interactive dashboard and reporting
DAX	KPI and calculated measures

6. Data Pre-Processing (Python)

Python was used for the complete data-cleaning and preprocessing stage before the data was loaded into Power BI.

6.1 Initial Data Quality Check

- Raw dataset shape: 10,000 rows $\times$ 15 columns.
- Duplicate rows: 0.
- Duplicate order IDs: 0.
- Missing customer_id: 200.
- Missing product_id: 200.
- Missing price: 200.
- Missing quantity: 200.
- Missing categorical columns category, delivery_status, Payment_method, device_type, channel each : 300.

6.2 Missing Value Handling

Categorical fields including category, delivery_status, payment_method, device_type, channel and customer_segment were filled with 'Unknown'. Price and quantity were filled using median values. Customer_id and product_id were not imputed because creating artificial identifiers could create false identities.

Before Cleaning

	0
order_id	0
customer_id	200
product_id	200
category	300
price	200
quantity	200
order_date	0
shipping_date	0
delivery_status	300
payment_method	300
device_type	300
channel	300
shipping_address	0
billing_address	0
customer_segment	300
dtype: int64	

After Cleaning

order_id	0
customer_id	200
product_id	200
category	0
price	0
quantity	0
order_date	0
shipping_date	0
delivery_status	0
payment_method	0
device_type	0
channel	0
shipping_address	0
billing_address	0
customer_segment	0
dtype: int64	

6.3 Categorical Standardization

Field	Inconsistent Values	Standardised Values
Category	book, electronic, home & living, clothings	Books, Electronics, Home, Clothing
Delivery Status	complete, awaiting	Delivered, Pending
Payment Method	debitcard, creditcard, pay pal	Debit Card, Credit Card, PayPal
Device Type	phone	Mobile
Customer Segment	v.i.p, repeat, new customer	VIP, Returning, New

Before Standardization

	count
category	
home	1652
beauty	1625
electronics	1622
toys	1619
clothing	1558
books	1557
unknown	300
book	18
clothings	18
electronic	16
home & living	15

After Standardization

	count
category	
home	1667
electronics	1638
beauty	1625
toys	1619
clothing	1576
books	1575
unknown	300
dtype: int64	

6.4 Date Conversion and Feature Engineering

```
df_clean["order_date"] = pd.to_datetime(df_clean ["order_date"])
df_clean ["shipping_date"] = pd.to_datetime(df_clean ["shipping_date"])

df["revenue"] = df_clean ["price"] * df_clean ["quantity"]
df["shipping_days"] = (df_clean ["shipping_date"] - df_clean ["order_date"]).dt.days
```

Before and After Data type Conversion

order_date	object
shipping_date	object

order_date	datetime64[ns]
shipping_date	datetime64[ns]

New Columns

	price	quantity	revenue
0	45.95	4.0	183.80
1	403.17	3.0	1209.51
2	317.45	2.0	634.90
3	24.08	3.0	72.24
4	494.90	1.0	494.90

6.5 Outlier Review

IQR-based screening identified potential price and quantity outliers. These records were retained because they may represent legitimate high-value or bulk orders. They were not deleted.

7. Data Modelling and DAX (Power BI)

Since the dataset was imported as a single fact table containing all required business attributes, ***no additional dimension tables or relationships were required.***

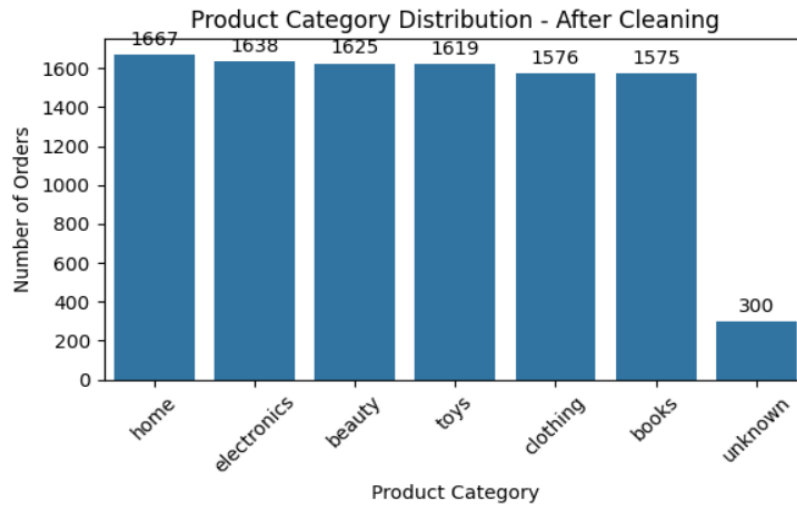
The cleaned e-commerce dataset was imported into Power BI as a single analytical table. The dashboard uses DAX measures for the main KPIs.

Measure	Formula
Total Revenue	SUM(ECommerce[revenue])
Total Orders	DISTINCTCOUNT(ECommerce[order_id])
Average Order Value	DIVIDE([Total Revenue],[Total Orders],0)
Average Shipping Days	AVERAGE(ECommerce[shipping_days])
Delivered Orders	CALCULATE([Total Orders], ECommerce[delivery_status] = "delivered")
Delivery Rate	DIVIDE([Delivered Orders],[Total Orders],0)

8. Analysis and Visualizations (Python)

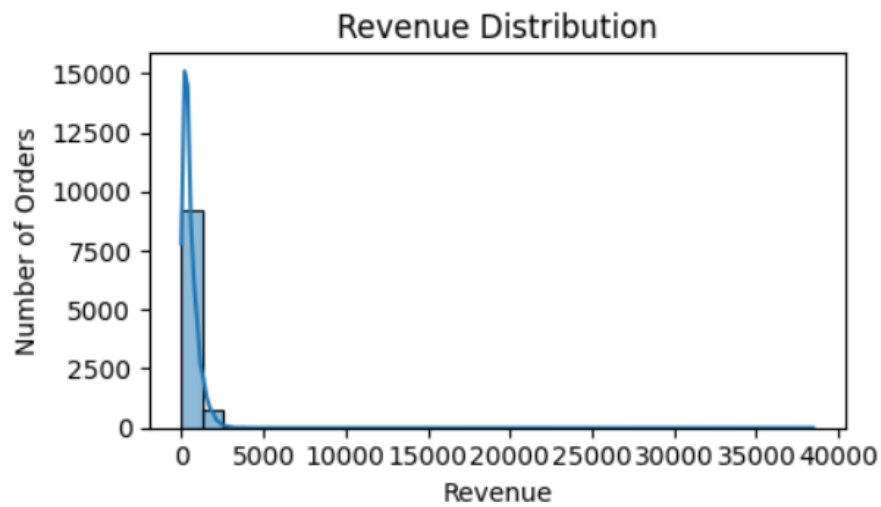
The following analytical visualizations were created in Python using the cleaned dataset. These Python visuals are used for the project analysis, while Power BI is used for the final interactive dashboard.

1. Product Category Distribution.

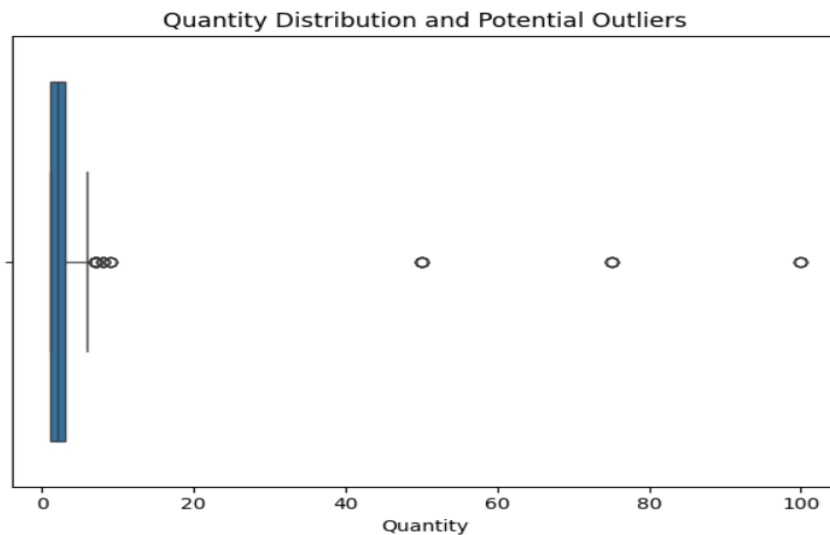


2. Revenue

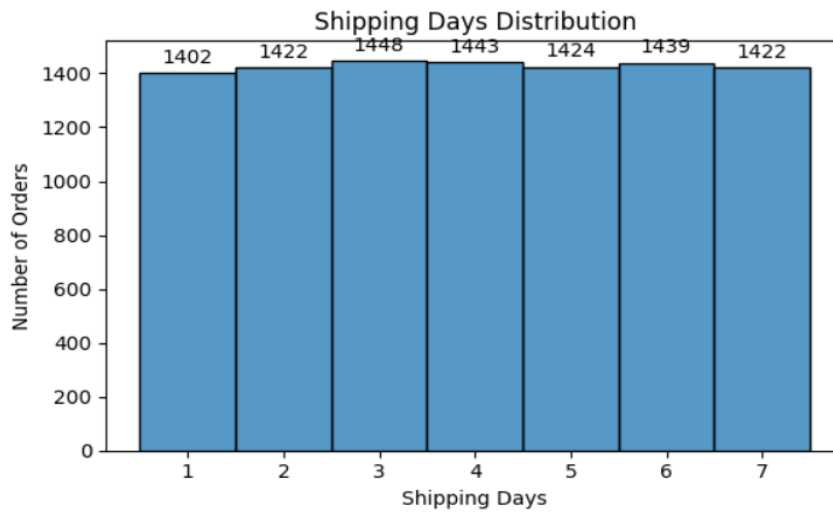
Distribution.



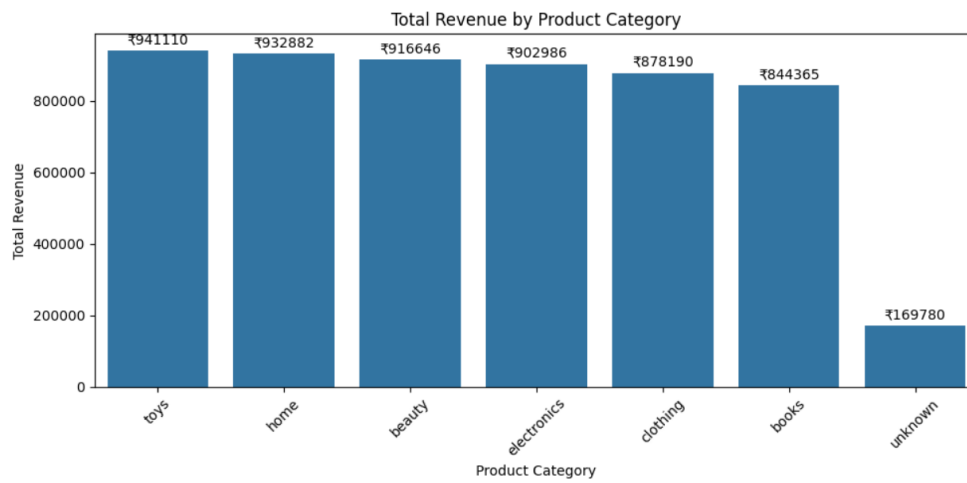
3. Quantity Distribution.



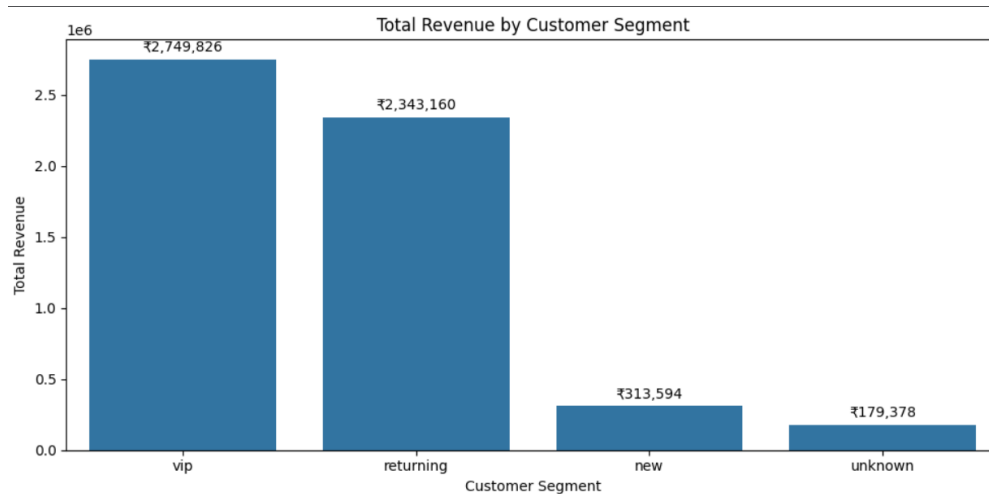
4. Shipping Days Distribution.



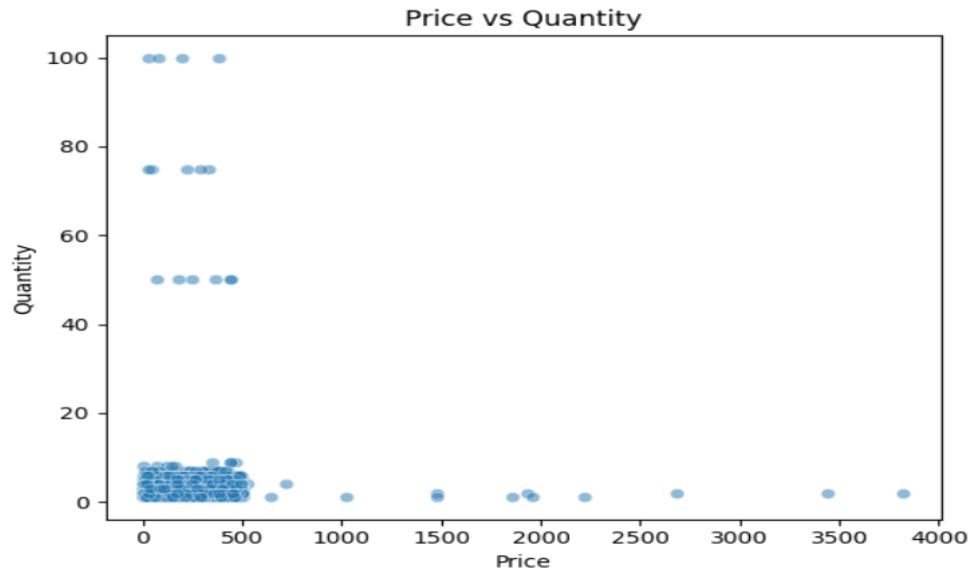
5. Revenue by Product Category.



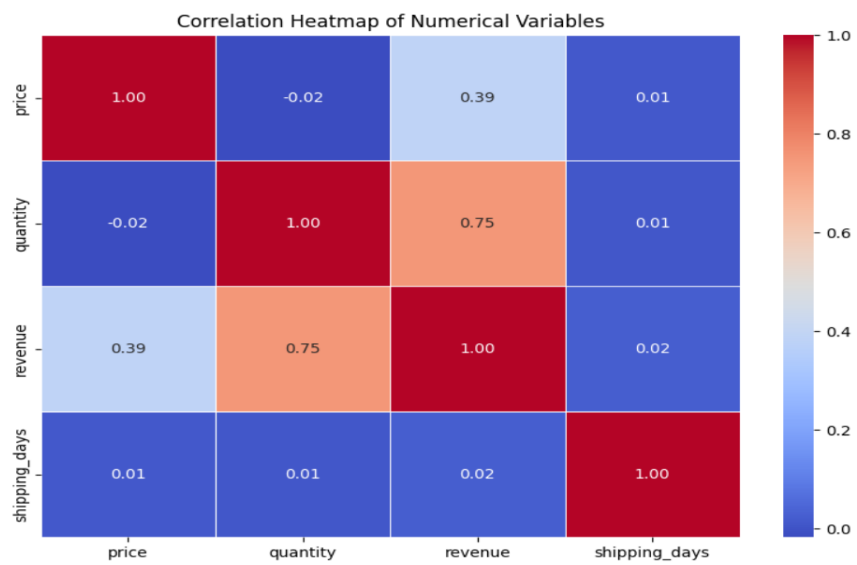
6. Revenue by Customer Segment.



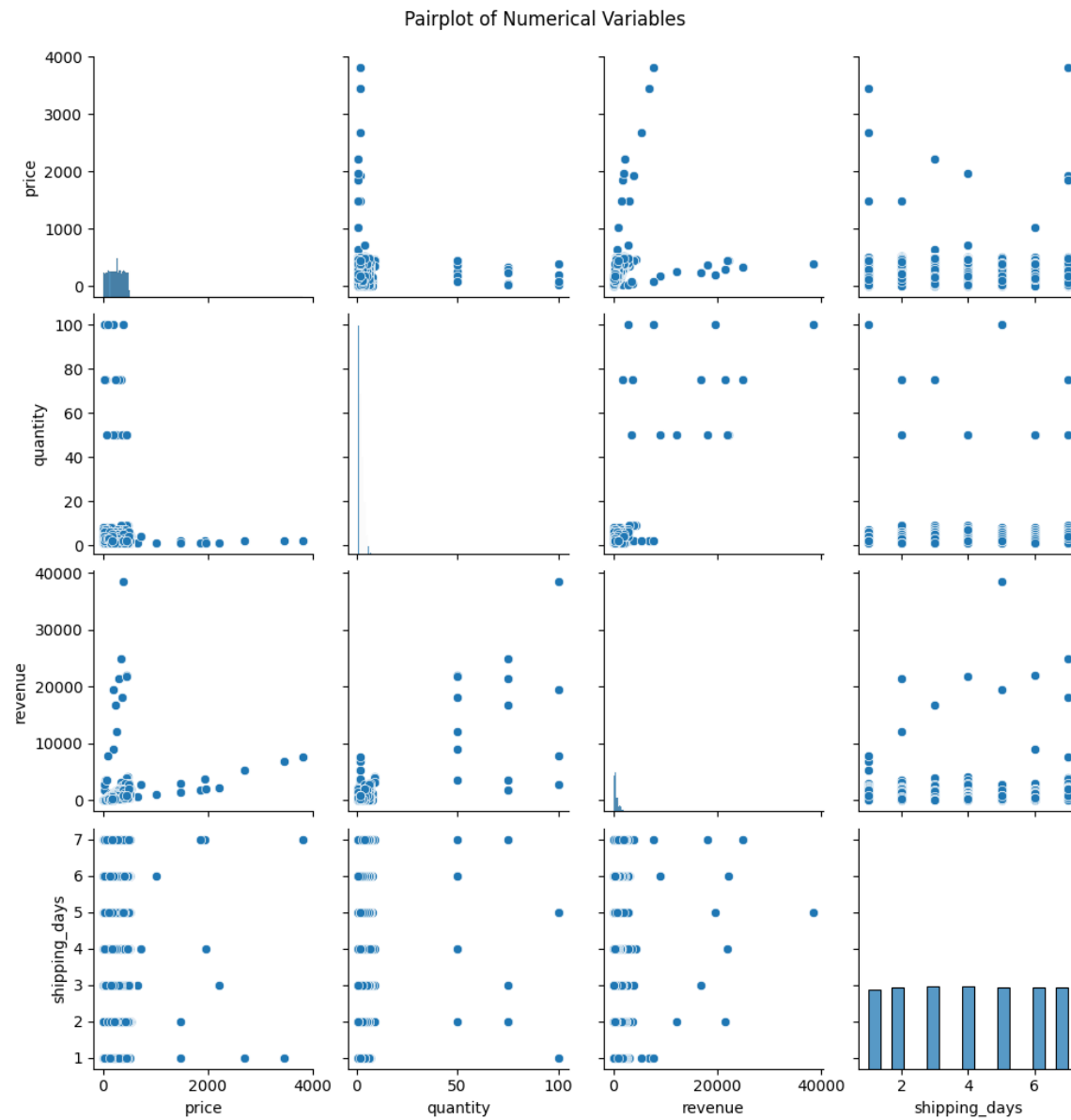
7. Price vs Quantity



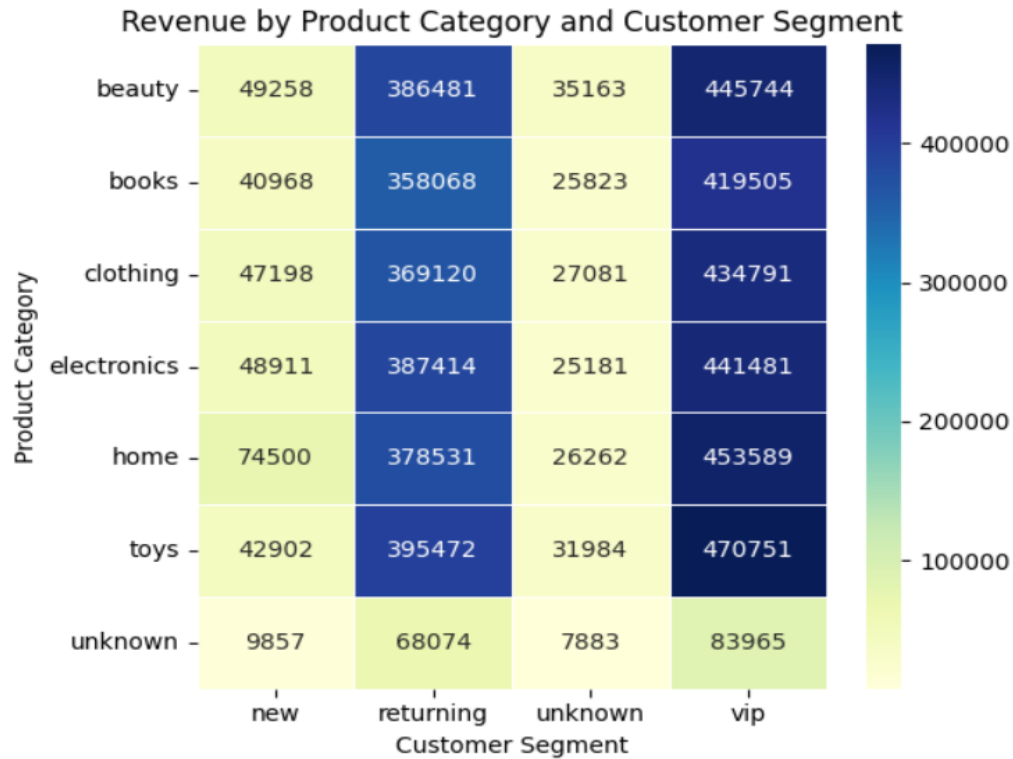
8. Correlation Heat map



9. Pair plot of Numerical Variables 'price', 'quantity', 'revenue', 'shipping days'



10. Revenue by Category and Customer Segment - Pivot + Heat map



8.1 Main Python Findings

- Revenue is right-skewed, with a relatively small number of high-value orders contributing strongly to total revenue.
- Toys is the highest-revenue product category at approximately \$0.94M.
- VIP customers generate the highest revenue at approximately \$2.7M.
- The price-versus-quantity relationship is very weak, indicating that higher-priced products do not necessarily have higher quantities.
- Average shipping time is approximately 4.01 days.
- Monthly revenue varies across the project period and should be monitored for business planning.

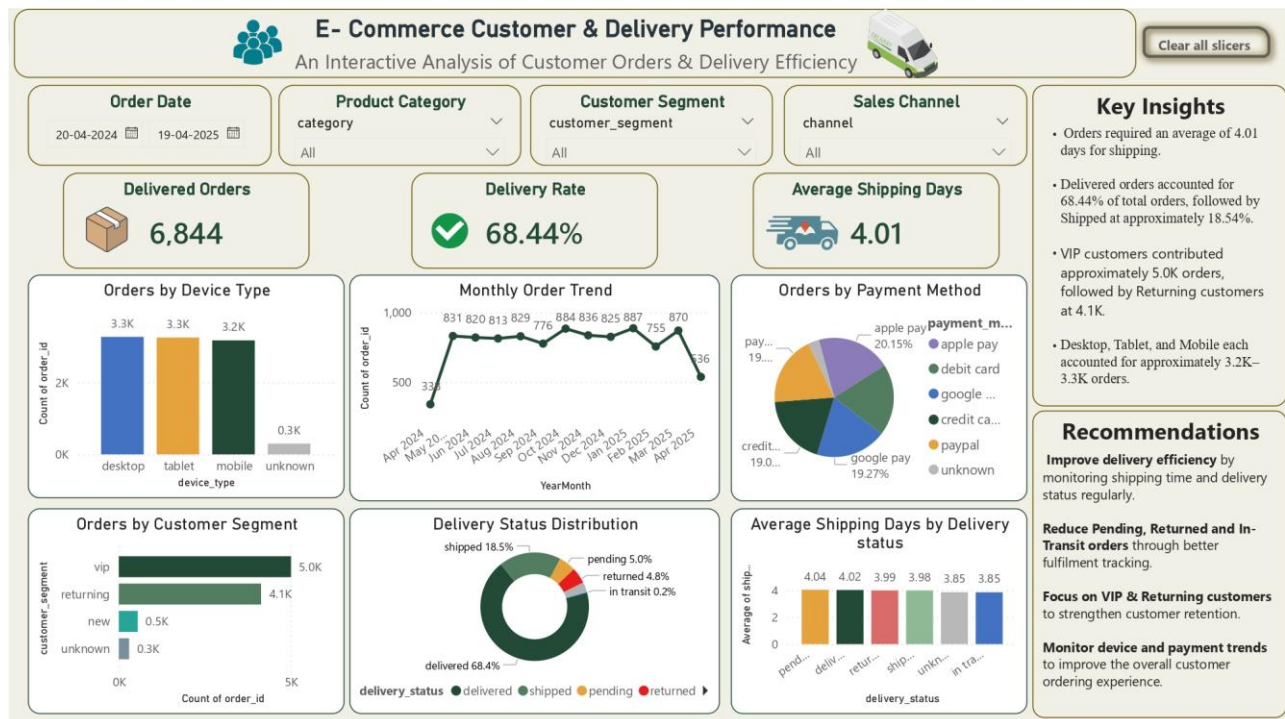
8.2 Power BI Dashboard – Sales & Revenue Performance

The updated Power BI dashboard contains KPI cards for Average Order Value, Total Revenue and Total Orders. It also includes slicers for Product Category, Customer Segment, Order Date and Sales Channel, together with revenue analysis by category, monthly revenue trend, device type, customer segment, payment method and sales channel.



8.3 Power BI Dashboard – Customer & Delivery Performance

The second dashboard contains KPI cards for Delivered Orders, Delivery Rate and Average Shipping Days. It includes order analysis by device type, customer segment and payment method, monthly order trend, delivery-status distribution and average shipping days by delivery status.



9. Insights & Recommendations

9.1 Key Insights

- The dashboard shows approximately \$5.59M revenue across 10,000 orders.
- Average Order Value is \$558.60.
- Toys generated the highest category revenue at approximately \$0.94M, followed by Home and Beauty.
- VIP customers generated the highest revenue at approximately \$2.7M, followed by Returning customers at \$2.3M.
- Social generated the highest sales-channel revenue at approximately \$1.43M.
- Orders required an average of 4.01 days for shipping.
- Delivered orders accounted for 68.44% of total orders, followed by Shipped at approximately 18.54%.
- VIP customers contributed approximately 5.0K orders, followed by Returning customers at approximately 4.1K.
- Desktop, Tablet and Mobile each contributed approximately 3.2K–3.3K orders.

9.2 Descriptive Insights

- Toys is the leading revenue category.
- VIP and Returning customers are the major revenue-contributing segments.
- Social is the highest-revenue sales channel in the dashboard.
- Delivery status is dominated by Delivered orders.
- Average shipping duration is close to four days.

9.3 Diagnostic Insights

- Revenue is concentrated more strongly in established customer segments than in New customers.
- The relatively balanced device distribution suggests that customers use multiple device types rather than depending on a single device.
- The delivery-status distribution shows that a significant portion of orders are still Shipped, Pending, Returned or In Transit, so delivery efficiency should continue to be monitored.
- Unknown values represent missing source information and should be reduced at data-entry/source level.

9.4 Predictive Insights

- Improving retention of VIP and Returning customers can support future revenue growth.
- High-revenue categories can be prioritised for inventory and promotional planning.
- Monthly revenue and order trends can help identify stronger and weaker business periods.
- Continuous delivery monitoring can help identify operational changes over time.

9.5 Recommendations

- Prioritise VIP and Returning customers through loyalty offers and personalised recommendations.
- Maintain sufficient inventory for high-revenue categories such as Toys and Home.
- Develop campaigns to convert New customers into Returning customers.
- Evaluate Social channel performance together with marketing cost before increasing campaign spending.
- Monitor Delivery Rate and Average Shipping Days regularly.
- Reduce Unknown values by improving data capture and standardisation at source.
- Add cost, discount and profit fields in a future version to extend the analysis from revenue performance to profitability.

10. Conclusions

The project successfully demonstrates an end-to-end e-commerce analytics workflow using Python and Power BI. Python was used for data cleaning, preprocessing, feature engineering and exploratory analysis, while Power BI was used to create interactive dashboards for sales, customer and delivery performance.

The updated dashboard reports approximately \$5.59M in revenue from 10,000 orders, an Average Order Value of \$558.60, an Average Shipping Days value of 4.01 days and a Delivery Rate of 68.44%. The analysis highlights Toys as the leading category, VIP customers as the strongest revenue segment and Social as the leading revenue channel.

Overall, the project provides a clear view of e-commerce sales and operational performance and supports data-driven decisions related to customers, products, channels and delivery operations.